

Unit 108 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 63.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 344 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 2.6 %)	Type	Sierran mixed conifer
Trees per acre	68	CWHR size / density	5 / D (cover 91 %)
QMD	24.5 in	SDI	274 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	222 sq ft/ac 24.5 in	BA QMD	0 sq ft/ac 0.0 in
TPA	68	TPA	0
Volume	52.3 MBF/ac (9501 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~164 sq ft/ac BA	Sampling error	5.4 % at 95 % (t = 2.04, n = 31)
Species (BA)	WF 68, PP 66, IC 36, DF 29, SP 22	Stratum standard	40 % - MEETS; 31 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	4	3.6	2.6	81	0.4
12-15.9	22	13.2	14.2	425	2.3
16-19.9	33	12.0	21.3	744	4.1
20-23.9	53	12.7	34.2	1310	7.2
24-27.9	60	10.7	38.7	1582	8.7
28-31.9	44	5.9	28.4	1227	6.7
32-35.9	39	4.0	25.2	1152	6.3
36-39.9	35	2.9	22.6	1100	6.1
40-43.9	26	1.7	16.8	860	4.7
44-47.9	6	0.3	3.9	188	1.0
48-51.9	7	0.3	4.5	250	1.4
52-55.9	6	0.2	3.9	230	1.3
56-59.9	5	0.2	3.2	186	1.0
60-63.9	4	0.1	2.6	166	0.9
All	344	68.1	221.9	9501	52.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	106	22.5	68.4	23.6	2863	15.7
PP	103	19.0	66.5	25.3	2891	15.9
IC	56	12.9	36.1	22.7	1471	8.1
DF	45	7.0	29.0	27.6	1290	7.1
SP	34	6.7	21.9	24.4	986	5.4
All	344	68.1	221.9	24.5	9501	52.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 108: plots on the 300 ft grid (orange = QA flag)

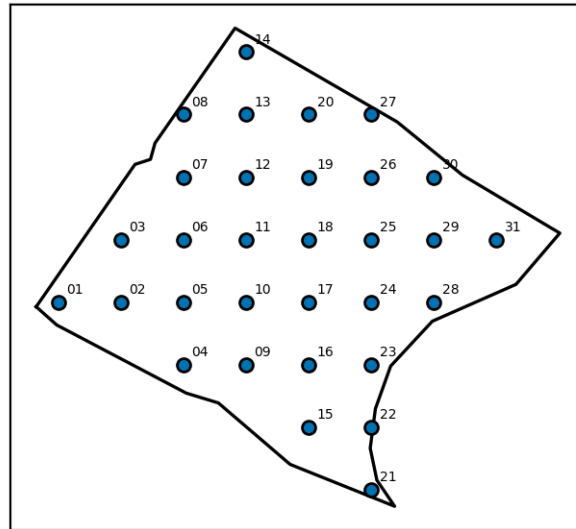


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
108-01	10	200	32	ok
108-02	11	220	71	ok
108-03	9	180	69	ok
108-04	10	200	65	ok
108-05	10	200	37	ok
108-06	12	240	69	ok
108-07	11	220	81	ok
108-08	11	220	198	ok
108-09	11	220	82	ok
108-10	12	240	51	ok
108-11	11	220	69	ok
108-12	14	280	71	ok
108-13	6	120	99	ok
108-14	13	260	16	ok
108-15	14	280	64	ok
108-16	10	200	64	ok
108-17	11	220	29	ok
108-18	11	220	14	ok
108-19	9	180	133	ok
108-20	12	240	197	ok
108-21	10	200	74	ok
108-22	12	240	40	ok
108-23	11	220	81	ok
108-24	12	240	37	ok
108-25	10	200	43	ok
108-26	13	260	48	ok
108-27	13	260	243	ok
108-28	11	220	16	ok
108-29	11	220	50	ok
108-30	10	200	67	ok
108-31	13	260	33	ok